Lecture 4

Duality: cups, caps and quantum dimension

Objectives. *A particle has an antiparticle, and a pair can be created from the vacuum and annihilated back into it. In a monoidal category this is a* dual*: an object X^* with a cup $1 \to X \otimes X^*$ and a cap $X^* \otimes X \to 1$ that cancel in a zigzag. This lecture gives the axioms, proves that they determine the dual, bends wires with them (the adjunction of Lecture 2, made literal), and closes loops to get traces and quantum dimensions.*

By the end of this lecture you will be able to:

- State the snake equations, prove that duals are unique up to a unique isomorphism, and compute f^* as a rotated box.
- Bend wires to prove $\mathrm{Hom}(U \otimes V, W) \cong \mathrm{Hom}(U, W \otimes V^*)$, and read teleportation as one snake.
- Define a pivotal structure, the trace and the quantum dimension, and prove cyclicity and multiplicativity by pictures.
- Show that for spin-$\frac{1}{2}$ with the singlet cup the snake is straight only with the cap $-\Omega^\dagger$, which closes the loop to -2, a pivotal choice, and identify the sign as a Frobenius–Schur indicator.

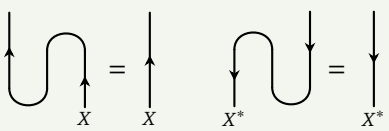
4.1 Duals and the snake

By coherence we work as if C were strict. A strand pointing up is an object X; a strand pointing down is its dual.

Definition 4.1 (Duals)**.** A *left dual* of X is an object X^* with morphisms

$$\mathrm{ev}_X : X^* \otimes X \to 1,$$
$$\mathrm{coev}_X : 1 \to X \otimes X^*,$$

drawn as a cap and a cup, such that the two *snakes* are straight:



(4.1)

that is, $(\mathrm{id}_X \otimes \mathrm{ev}_X) \circ (\mathrm{coev}_X \otimes \mathrm{id}_X) = \mathrm{id}_X$ and $(\mathrm{ev}_X \otimes \mathrm{id}_{X^*}) \circ (\mathrm{id}_{X^*} \otimes \mathrm{coev}_X) = \mathrm{id}_{X^*}$. A *right dual* ${}^*X$ has $\mathrm{ev}'_X : X \otimes {}^*X \to 1$ and $\mathrm{coev}'_X : 1 \to {}^*X \otimes X$ with the mirror-image snakes. C is *rigid* if every object has both.

Heunen–Vicary and Selinger write A^* for the *right* dual; ours is EGNO's.

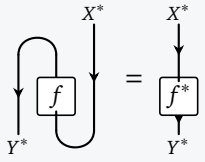
Theorem 4.2 (Duals are unique; the rotated box · *proved given coherence*)**.**

(a) If $(X^*, \mathrm{ev}, \mathrm{coev})$ *and* $(X', \mathrm{ev}', \mathrm{coev}')$ *are left duals of* X*, then*

$$\varphi = (\mathrm{ev} \otimes \mathrm{id}_{X'}) \circ (\mathrm{id}_{X^*} \otimes \mathrm{coev}')$$

is the unique isomorphism $X^* \to X'$ *with* $\mathrm{ev}' \circ (\varphi \otimes \mathrm{id}_X) = \mathrm{ev}$.

(b) For $f \colon X \to Y$ *the* dual morphism $f^* \colon Y^* \to X^*$*, the box rotated by a half turn,*



(4.2)

satisfies $(g \circ f)^* = f^* \circ g^*$ *and* $\mathrm{id}^* = \mathrm{id}$*: the dual is a contravariant functor.*

Proof. (a) First expand the compatibility equation. By interchange and the primed snake,

$$\begin{aligned} &\mathrm{ev}' \circ (\varphi \otimes \mathrm{id}_X) \\ &\quad = \mathrm{ev} \circ \big(\mathrm{id}_{X^*} \otimes (\mathrm{id}_X \otimes \mathrm{ev}')\big) \\ &\qquad \circ (\mathrm{id}_{X^*} \otimes \mathrm{coev}' \otimes \mathrm{id}_X) \\ &\quad = \mathrm{ev}\,. \end{aligned}$$

Let $\psi = (\mathrm{ev}' \otimes \mathrm{id}_{X^*}) \circ (\mathrm{id}_{X'} \otimes \mathrm{coev})$. Using this compatibility,

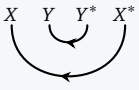
$$\begin{aligned} \psi \circ \varphi &= (\mathrm{ev} \otimes \mathrm{id}_{X^*}) \circ (\mathrm{id}_{X^*} \otimes \mathrm{coev}) \\ &= \mathrm{id}_{X^*}, \end{aligned}$$

by the other snake. Likewise $\varphi \circ \psi = \mathrm{id}_{X'}$. Any φ' satisfying the cap compatibility equation equals $(\mathrm{ev} \otimes \mathrm{id}) \circ (\mathrm{id} \otimes \mathrm{coev}')$ after inserting a snake, which is φ.

(b) Stack the pictures of f^* and g^*; the cap of one and the cup of the other form a snake, which straightens to give the picture of $(g \circ f)^*$. ∎

Proposition 4.3 (The dual of a tensor product · *proved given coherence*)**.** *If* X^* *and* Y^* *are left duals of* X *and* Y*, then* $Y^* \otimes X^*$*, in that order, is a left dual of* $X \otimes Y$*, with the nested cup and cap*

$$\mathrm{coev}_{X \otimes Y} =$$



(4.3)

and the corresponding nested cap.

Proof. Each snake for $X \otimes Y$ is a snake for Y inside a snake for X; straighten the inner one, then the outer one. ∎

The order reverses, as for the inverse of a product of matrices, and the cups nest rather than cross.

Example 4.4 (Duals)**.**

- $\mathrm{Vect}_{\mathrm{fd}}$: X^* is the dual space, $\mathrm{ev}(\varphi \otimes v) = \varphi(v)$, and $\mathrm{coev}(1) = \sum_i e_i \otimes e^i$ for any basis (e_i) and its dual basis; it is the identity of X under $\mathrm{Hom}(X, X) \cong X \otimes X^*$, so it does not depend on the basis. The rotated box is the transpose, the dual map of Lecture 1.

- $\mathrm{Rep}(G)$: the dual is the contragredient representation, $g \mapsto (\rho(g)^{-1})^*$ (*Symmetries* Prop. 3.10); ev and coev are those of $\mathrm{Vect}_{\mathrm{fd}}$, and are intertwiners.
- Vec_G^{ω}: $\delta_g^* = \delta_{g^{-1}}$. With $\mathrm{coev}_g = \mathrm{id}$ and the unitors identities, the first snake equals $\mathrm{ev}_g\ \omega(g, g^{-1}, g)$, from the associator in the middle; so $\mathrm{ev}_g = \omega(g, g^{-1}, g)^{-1}$ id. The second snake then holds by the cocycle condition. For the $\mathbb{Z}_2$ category of Lecture 3 with $\omega(s,s,s) = -1$, the naive choice ev = coev = id gives a snake equal to $-\mathrm{id}_s$.

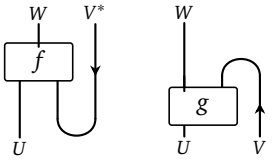
4.2 Bending wires; teleportation as one picture

Theorem 4.5 (Bending wires · *proved given coherence*). *For a left dual V^* of V, the maps*

$$f \mapsto (f \otimes \mathrm{id}_{V^*}) \circ (\mathrm{id}_U \otimes \mathrm{coev}_V),$$
$$g \mapsto (\mathrm{id}_W \otimes \mathrm{ev}_V) \circ (g \otimes \mathrm{id}_V)$$

are inverse bijections $\mathrm{Hom}(U \otimes V, W) \cong \mathrm{Hom}(U, W \otimes V^*)$, natural in U and W.

Proof. In pictures the first map bends the V strand of f up to the right, and the second bends it back down:



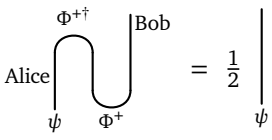
(4.4)

Bending up and then down leaves a snake on the V strand, which straightens by (4.1); the other order straightens the V^* snake. Naturality holds because boxes on U and W slide past the cup and cap. ∎

So $- \otimes V$ is left adjoint to $- \otimes V^*$. In $\mathrm{Vect}_{\mathrm{fd}}$, where $\mathrm{Hom}(V,W) \cong W \otimes V^*$ (*Linear Algebra* Prop. 12.18), this is Lecture 2's adjunction

$$\mathrm{Hom}(U \otimes V, W) \cong \mathrm{Hom}(U, \mathrm{Hom}(V, W)).$$

Teleportation. In $\mathrm{Hilb}_{\mathrm{fd}}$ with an orthonormal basis, coev becomes $\sum_i |ii\rangle = \sqrt{2}\,|\Phi^+\rangle$ for a qubit, after identifying X^* with X by the basis (*Quantum Computing*, L4). Alice holds an unknown $|\psi\rangle$ and half of $|\Phi^+\rangle$; Bob holds the other half. If Alice's Bell measurement returns Φ^+, the picture is the snake: the cup creates the pair, Alice's cap closes her two strands, and Bob's strand carries $|\psi\rangle$, with amplitude $\frac{1}{2}$ from the two normalisations,



(4.5)

that is,

$$(\langle\Phi^+| \otimes \mathrm{id})(|\psi\rangle \otimes |\Phi^+\rangle) = \tfrac{1}{2}\,|\psi\rangle\,.$$

The other three outcomes put a Pauli matrix on Bob's strand, which he undoes.

4.3 Pivotal, trace, dimension, spherical

To close a loop we need a cap on $X \otimes X^*$, the other way round. In a rigid category X^{**} is a left dual of X^*, so we need to compare X with X^{**}.

Definition 4.6 (Pivotal structure, trace, dimension)**.** A *pivotal structure* is a natural isomorphism $a_X : X \to X^{**}$ with $a_{X\otimes Y} = a_X \otimes a_Y$ (under $(X\otimes Y)^{**} \cong X^{**}\otimes Y^{**}$). It gives a second cap and cup,

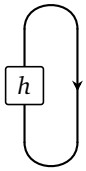
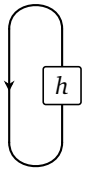
$$\widetilde{\mathrm{ev}}_X : X \otimes X^* \to 1,$$
$$\widetilde{\mathrm{ev}}_X = \mathrm{ev}_{X^*} \circ (a_X \otimes \mathrm{id}_{X^*}),$$
$$\widetilde{\mathrm{coev}}_X : 1 \to X^* \otimes X,$$

the second defined likewise from coev_{X^*} and a_X^{-1}; so X^* is also a right dual. For $h : X \to X$ the *left* and *right traces* close the strand to the right and to the left,

$$\mathrm{tr}_L(h) = \widetilde{\mathrm{ev}}_X \circ (h \otimes \mathrm{id}_{X^*}) \circ \mathrm{coev}_X,$$
$$\mathrm{tr}_R(h) = \mathrm{ev}_X \circ (\mathrm{id}_{X^*} \otimes h) \circ \widetilde{\mathrm{coev}}_X,$$

elements of $\mathrm{End}(1)$. The *quantum dimension* is $d_X = \mathrm{tr}_L(\mathrm{id}_X)$, a closed loop. The structure is *spherical* if $\mathrm{tr}_L = \mathrm{tr}_R$.

In pictures the two traces close the strand on opposite sides:

$\mathrm{tr}_L(h) =$  $\qquad \mathrm{tr}_R(h) =$  (4.6)

In $\mathrm{Vect}_{\mathrm{fd}}$ the canonical a_V of Lecture 1 is a pivotal structure, both traces are the ordinary trace, and $d_V = \dim V$:

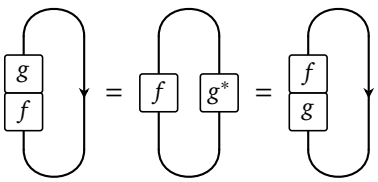
 $= d_X.$ (4.7)

Theorem 4.7 (Traces · *proved given coherence*)**.** *In a rigid $\mathbb{C}$-linear category with bilinear $\otimes$, $\mathrm{End}(1) = \mathbb{C}$ and a pivotal structure, for $f : X \to Y$, $g : Y \to X$ and $h : X \to X$, $h' : X' \to X'$,*

$$\mathrm{tr}_L(g \circ f) = \mathrm{tr}_L(f \circ g),$$
$$\mathrm{tr}_L(h \otimes h') = \mathrm{tr}_L(h)\,\mathrm{tr}_L(h'),$$

so $d_{X\otimes Y} = d_X\, d_Y$ and $d_1 = 1$; likewise for tr_R.

Proof. Cyclicity: slide g around the loop. Rotated by the cup and the cap it becomes g^* on the down-going strand, by (4.2) and the naturality of a, and slides back up on the other side:

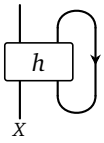
 (4.8)

Multiplicativity: $a_{X\otimes X'} = a_X \otimes a_{X'}$ makes the loop on $X \otimes X'$ two nested loops, one around each factor, and the inner one, a number, can be moved out of the picture (Lecture 3). ∎

In $\mathrm{Rep}(G)$ with its canonical pivotal structure, a closed loop gives $\dim V$. For the simple representations of S_3 these values are 1, 1 and 2.

Closing only the right strand of $h\colon X\otimes Y\to X\otimes Y$ is the *partial trace* over Y, the operation of *QM-I* (Def. 11.2) that gives the state of a subsystem:

$$\mathrm{tr}_Y(h) \;=\; \tag{4.9}$$



Dimensions need not be positive integers. On Vec_G for $G=\mathbb{Z}_2$, the choice $a_s=-\mathrm{id}$ is also a pivotal structure, and it gives $d_s=-1$; still $d_{s\otimes s}=d_s^2=1=d_1$, as the theorem requires. The pivotal structure is a choice, and the quantum dimension depends on it. A character χ of G gives one on Vec_G with $d_g=\chi(g)$ (EGNO Ex. 4.7.16). For $G=\mathbb{Z}_3=\{1,r,r^2\}$ and $\chi(r)=e^{2\pi i/3}$,

$$d_{r^*}=d_{r^2}=e^{4\pi i/3}\neq e^{2\pi i/3}=d_r,$$

so this pivotal structure is not spherical. A spherical one has $d_X=d_{X^*}$: the right trace of id_X, the loop drawn the other way round, is the left trace of id_{X^*} (EGNO Prop. 4.7.3, Ex. 4.7.9), so $\mathrm{tr}_L=\mathrm{tr}_R$ gives $d_X=d_{X^*}$.

4.4 The Penrose sign and the FS indicator

Let $V=\mathbb{C}^2$ be spin-$\frac12$. The SU(2)-invariant form Ω, with $\Omega(e_1,e_2)=1=-\Omega(e_2,e_1)$, identifies V with V^*, and as a cup

$$\Omega\colon\mathbb{C}\to V\otimes V,\qquad 1\mapsto e_1\otimes e_2-e_2\otimes e_1,$$

which is $\sqrt{2}$ times the singlet of *Symmetries* (L14). It is an intertwiner, since $(U\otimes U)\Omega=\det(U)\,\Omega=\Omega$ for $U\in\mathrm{SU}(2)$.

Theorem 4.8 (The Penrose sign). *A cap $c\colon V\otimes V\to\mathbb{C}$ satisfies the snake equation with the cup Ω if and only if $c=-\Omega^\dagger$. Then both snakes are straight and the loop is $c\circ\Omega=-2$. With the cap $\Omega^\dagger$ instead, the snake is $-\mathrm{id}$ and the loop is $+2$.*

Proof. Write $c(e_i\otimes e_j)=c_{ij}$. The snake $(\mathrm{id}\otimes c)(\Omega\otimes\mathrm{id})$ sends e_k to $e_1c_{2k}-e_2c_{1k}$, so it is the identity exactly when $c_{21}=1$, $c_{12}=-1$ and $c_{11}=c_{22}=0$; that is $c=-\Omega^\dagger$, since $\Omega^\dagger(e_1\otimes e_2)=1=-\Omega^\dagger(e_2\otimes e_1)$. The loop is $-\Omega^\dagger\Omega=-\|\Omega\|^2=-2$, and the other snake is checked the same way. With $c=\Omega^\dagger$ every sign flips. ∎

Keep the left-dual cup Ω and cap $-\Omega^\dagger$. The closing cap $\widetilde{\mathrm{ev}}_V=\mathrm{ev}_{V^*}\circ(a_V\otimes\mathrm{id})$ also depends on the pivotal structure. The canonical a of $\mathrm{Vect}_{\mathrm{fd}}$ gives the closing cap $\Omega^\dagger$, so $d_V=2$. Replacing a by $(-1)^{2j}a$ on spin j changes this closing cap to $-\Omega^\dagger$ on V, giving -2 (Sheet 4). The convention "cap = cup†" (*Quantum Computing*, C12) uses the canonical closing cap; the straight Penrose cap uses the changed pivotal structure.

Definition 4.9 (Frobenius–Schur indicator). Let X be a simple object in a pivotal category with $\mathrm{End}(X)=\mathbb{C}$, and $j\colon X\to X^*$ an isomorphism. Then $j^*\circ a_X\colon X\to X^*$ is another, so it defines the *Frobenius–Schur indicator* κ_X by

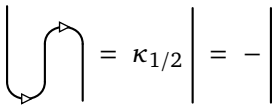
$$j^*\circ a_X=\kappa_X\, j.$$

Applying the dual again gives, by naturality of a and $a_{X^*} = (a_X^*)^{-1}$ (EGNO Ex. 4.7.9),

$$\begin{aligned}\kappa_X^2 j &= (j^* \circ a_X)^* \circ a_X \\ &= a_X^* \circ j^{**} \circ a_X \\ &= a_X^* \circ a_{X^*} \circ j = j.\end{aligned}$$

Thus $\kappa_X = \pm 1$. Rescaling j leaves this sign unchanged, since the dual is linear.

For spin-$\frac{1}{2}$, $j(v) = \Omega(\cdot, v)$ has the antisymmetric matrix $\left(\begin{smallmatrix} 0 & 1 \\ -1 & 0 \end{smallmatrix}\right)$, and j^* is its transpose, $-j$: so $\kappa_{1/2} = -1$. With the canonical pivotal structure on representations, an object with a symmetric invariant form has $\kappa = +1$. Bonderson's flags record the self-duality identification j used when bending a strand. Here the flagged cup is Ω and the same-way flagged cap is $\Omega^\dagger$. Reversing either flag multiplies its morphism by κ_X, so opposed flags give a straight snake. Same-way flags give



(4.10)

and the sign returns in Lecture 6 as an F-symbol, κ_a / d_a.

4.5 Dagger

A *dagger category* has an operation $f \mapsto f^\dagger$, reversing morphisms, with $(g \circ f)^\dagger = f^\dagger \circ g^\dagger$, $\mathrm{id}^\dagger = \mathrm{id}$ and $f^{\dagger\dagger} = f$; in $\mathrm{Hilb}_{\mathrm{fd}}$ it is the adjoint.

Theorem 4.10 ($\mathrm{Hilb}_{\mathrm{fd}}$ is dagger pivotal · *sketched*)**.** *With X^* given the inner product for which dual bases of orthonormal bases are orthonormal, the caps of $\mathrm{Hilb}_{\mathrm{fd}}$ are the daggers of its cups: $\widetilde{\mathrm{ev}}_X = \mathrm{coev}_X^\dagger$ and $\mathrm{ev}_X = \widetilde{\mathrm{coev}}_X^\dagger$ (Heunen–Vicary Def. 3.51 and Prop. 3.52).*

In an orthonormal basis,

$$\begin{aligned}&\mathrm{coev}^\dagger(e_j \otimes e^k) \\ &\quad = \left\langle \sum_i e_i \otimes e^i, e_j \otimes e^k \right\rangle \\ &\quad = \delta_{jk}.\end{aligned}$$

Spin-$\frac{1}{2}$ does not contradict it: there the cup is Ω, which identifies V^* with V by an antisymmetric form, not by the inner product.

Summary.

- *Handed on:* cups and caps with straight snakes; duals are unique up to a unique isomorphism, and f^* is the rotated box (Theorem 4.2).
- *Bending:* $\mathrm{Hom}(U \otimes V, W) \cong \mathrm{Hom}(U, W \otimes V^*)$ (Theorem 4.5); teleportation is a snake.
- *Loops:* a pivotal structure closes loops; traces are cyclic and multiplicative, and the quantum dimension d_X is a loop (Theorem 4.7).
- *The sign:* the straight Penrose cap is $-\Omega^\dagger$, with loop -2 (Theorem 4.8); the canonical pivotal structure has $\kappa_{1/2} = -1$.
- *Next:* Lecture 5 fuses particles: semisimple rigid categories with finitely many simple objects, and their fusion rules.

References

Etingof, Gelaki, Nikshych and Ostrik, *Tensor Categories*: duals, Defs. 2.10.1–2.10.2 and Prop. 2.10.5, pp. 40–41; examples, Exs. 2.10.12–2.10.14, pp. 42–43; bending, Prop. 2.10.8, p. 42; traces, pivotal structures and dimensions, Defs. 4.7.1, 4.7.7, 4.7.11, 4.7.14 and Thm. 4.7.15, pp. 73–75 (Ex. 2.10.14 prints $\omega(g, g^{-1}, g)$; the inverse is right). Heunen and Vicary, *Categories for Quantum Theory*: duals and snakes, Def. 3.1, pp. 89–90 (their A^* is our right dual); teleportation, §3.2, p. 99; pivotal, Def. 3.24, p. 107; dagger pivotal, Def. 3.51 and Prop. 3.52, p. 117. Kitaev, *Anyons in an exactly solved model and beyond*, cond-mat/0506438, App. E, pp. 74–77 (the Frobenius–Schur indicator). Bonderson, PhD thesis (Caltech, 2007), (2.22)–(2.27), pp. 20–21 (flags). Selinger, *A survey of graphical languages*, §4.1, p. 18. Kauffman and Lins, *Temperley–Lieb Recoupling Theory*, §§8.1–8.2, pp. 77–79 (the loop −2 as Penrose's theory).